



SECR1213-12 (Network Communications)

Group 5

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TABLE OF CONTENTS

NO.	TITLE	PAGE NUMBER
1.	Introduction	2
2.	Get the Network Address from Your Lecturer	3
3.	IP Assigation	4-7
	3.1 Network and Broadcast Address for Each Subnet	4
	3.2 IP Assigation for Each Work Area	6
4.	Reflection for Task 5	8
5.	Minute Meeting	9-10

1.0 Introduction

IP addressing plays a crucial role in designing and managing network infrastructure, ensuring smooth communication between devices while avoiding address conflicts. For our project, we were tasked with creating an IP addressing scheme tailored to the given network. This involves dividing the provided network address into subnets and assigning them to different work areas within the faculty, ensuring efficient allocation and utilization of IP addresses.

This task focuses on analyzing the network's requirements and structuring an appropriate IP addressing plan. Using subnetting techniques, we allocated sufficient IP ranges to each work area, such as the labs, student lounge, and video conferencing room, ensuring scalability for future expansions. The process emphasizes simplicity, organization and the prevention of address conflicts.

2.0 Get the Network Address from your lecturer

Group	Network Address
12.1	192.160.0.0/8
12.2	192.162.0.0/8
12.3	192.164.0.0/8
12.4	192.166.0.0/8
12.5	192.168.0.0/8
12.6	192.170.0.0/8

Table 1.0

Table 1.0 shows the list of network addresses of each group. The network address given to our group is 192.168.0.0/8. We assume that the subnet mask is /24 or 255.255.255.0

IP Address (decimal)	192.168.0.0
IP Address (binary)	11000000 . 10101000 . 00000000 . 00000000
Subnet Mask (decimal)	255 . 255 . 255 . 0
Subnet Mask (binary)	11111111 . 11111111 . 11111111 . 00000000

We divided the subnet mask into two parts: 24 bits for the network address and 8 bits for the host. Using the AND operation between the IP address and the subnet mask, we determined the network address to be 192.168.0.0 and the broadcast address to be 192.168.0.255. This provides a valid range of IP addresses from 192.168.0.1 to 192.168.0.254 for use within the building.

Network Address	Broadcast Address	Range of Usable Addresses
192.168.0.0	192.168.0.255	192.168.0.1 - 192.168.0.254

3.0 IP Assignment

In this section, the IP address allocation for the network is structured to ensure that each work area is assigned a unique subnet. This organization prevents addresses conflicts and ensures efficient IP usage across all devices and work areas.

3.1 Network and Broadcast Address for Each Subnet

Each subnet is designed to accommodate the devices in a specific work area. The subnets are created using the /24 subnet mask, providing sufficient IP addresses for current and future expansion while maintaining simplicity and scalability. The table below outlines the network portions, broadcast addresses, and usable IP ranges for each work area.

Subnet	Network Portion	Subnet Bits	Host Portion	Network Address	Broadcast Address
1	11000000 . 10101000	00000001	00000000	192.168.1.0	192.168.1.255
2	11000000 . 10101000	00000010	00000000	192.168.2.0	192.168.2.255
3	11000000 . 10101000	00000011	00000000	192.168.3.0	192.168.3.255
4	11000000 . 10101000	00000100	00000000	192.168.4.0	192.168.4.255
5	11000000 . 10101000	00000101	00000000	192.168.5.0	192.168.5.255
6	11000000 . 10101000	00000110	00000000	192.168.6.0	192.168.6.255
7	11000000 . 10101000	00000111	00000000	192.168.7.0	192.168.7.255
8	11000000 . 10101000	00001000	00000000	192.168.8.0	192.168.8.255

Work Area	Devices Count	Network Address	Broadcast Address	Usable IP Range	Subnet Mask
Hybrid Classroom	4	192.168.1.0	192.168.1.255	192.168.1.1 - 192.168.1.254	255.255.255.0/24
Cafe	3	192.168.2.0	192.168.2.255	192.168.2.1 - 192.168.2.254	255.255.255.0/24
Video Conferencing Room	11	192.168.3.0	192.168.3.255	192.168.3.1 - 192.168.3.254	255.255.255.0/24
Student Lounge	4	192.168.4.0	192.168.4.255	192.168.4.1 - 192.168.4.254	255.255.255.0/24
General Purpose Lab 1	39	192.168.5.0	192.168.5.255	192.168.5.1 - 192.168.5.254	255.255.255.0/24
General Purpose Lab 2	39	192.168.6.0	192.168.6.255	192.168.6.1 - 192.168.6.254	255.255.255.0/24
Cisco Network Lab	45	192.168.7.0	192.168.7.255	192.168.7.1 - 192.168.7.254	255.255.255.0/24
Embedded Lab	38	192.168.8.0	192.168.8.255	192.168.8.1 - 192.168.8.254	255.255.255.0/24

3.2 IP Assnigation for Each Work Area

This section details the specific IP address assignments for individual devices within each work area. The allocation ensures that every device has a unique IP address within its designated subnet, facilitating smooth network communication and preventing IP conflicts. The table below provides the assigned IP addresses for devices such as Smart TVs, PCs, monitors, and wireless access points in the respective work areas.

Work Area	Devices	IP Range
Ground Floor		
Hybrid Classroom	Smart TV	192.168.1.1 - 192.168.1.2
	Switch	192.168.1.3
	Wireless Access Point	192.168.1.4
Cafe	Smart TV	192.168.2.1
	Switch	192.168.2.2
	Wireless Access Point	192.168.2.3
Video Conferencing Room	Smart TV	192.168.3.1
	Monitors	192.168.3.2 - 192.168.3.8
	PC	192.168.3.9
	Switch	192.168.3.10
	Wireless Access Point	192.168.3.11
Student Lounge	Smart TV	192.168.4.1
	PC	192.168.4.2
	Switch	192.168.4.3
	Wireless Access Point	192.168.4.4

First Floor		
General Purpose Lab 1	PC	192.168.5.1 - 192.168.5.36
	Switch	192.168.5.37
	Wireless Access Point	192.168.5.38
	Server	192.168.5.39
General Purpose Lab 2	PC	192.168.6.1 - 192.168.6.36
	Switch	192.168.6.37
	Wireless Access Point	192.168.6.38
	Server	192.168.6.39
Cisco Network Lab	PC	192.168.7.1 - 192.168.7.41
	Switch	192.168.7.42
	Wireless Access Point	192.168.7.43
	Server	192.168.7.44
	Projector	192.168.7.45
Embedded Lab	PC	192.168.8.1 - 192.168.8.34
	Switch	192.168.8.35
	Wireless Access Point	192.168.8.36
	Server	192.168.8.37
	Projector	192.168.8.38

4.0 Reflection for Task 5

Task 5 provided valuable insights into designing and implementing an effective IP addressing scheme for a large network. By carefully analyzing the requirements of each work area, we were able to allocate unique subnets and assign IP addresses efficiently, ensuring smooth communication and avoiding conflicts. This process emphasized the importance of subnetting and understanding the role of network and host portions in IP address management.

Additionally, the task highlighted the significance of organization and scalability. The use of structured subnetting, such as the /24 subnet mask, allowed us to create a network that meets current needs while leaving room for future expansion. This experience reinforced the practical application of network theory and the importance of thoughtful planning in real-world scenarios.

5.0 Minute Meeting

DATE/TIME	25/1/2025 - 2100H		
LOCATION	Google Meet		
AGENDA	i) Discuss about task 5 requirements ii) Discuss regarding IP address assignation iii) Divide task amongst the group members		
Meeting MC	Muhammad Afiq Danish bin Mohd Hazni		
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Muhammad Afiq Danish bin Mohd Hazni		2100H	-
Muhammad Syahmi Faris bin Rusli		2100H	-
Muhammad Afiq Danial bin Rozaidie		2100H	-
Dheshieghan A/L Saravana Moorthy		2100H	-
MINUTES			
NO.	ITEMS DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss the number of devices in each floor and work areas	Every member list the number of devices used in their responsible work areas and floor	Afiq Danish, Afiq Danial, Syahmi, Dhesh
2	Discuss on what prefix to be used for subnet mask	Syahmi propose the idea to use /24 for IP address distribution	Syahmi
3	Discuss about who in charge for ground floor IP address distribution	Afiq Danish and Syahmi will in charge for ground floor	Afiq Danish, Syahmi
4	Discuss about who in charge for first floor IP address distribution	Afiq Danial and Dhesh will in charge for first floor	Afiq Danial, Dhesh

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MUHAMMAD SYAHMI FARIS BIN RUSLI A23CS0138 (You, presenting) Stop presenting

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TASK 5: IP ADDRESSING SCHEME (10 Marks)

IP addressing is crucial in making sure that every host can connect to the network without conflict of addresses. In this task, you need to explore how best to divide the subnetwork from the Network Address assigned to your group.

- Get the Network Address for your group as stated in the document.
- Divide it in the best possible way for your network - all the different labs and rooms.

Deliverables:

- Complete and detailed workings of your subnetting and IP assignment to each lab and room.
- Meeting minutes (all meeting minutes MUST be informational and specific).
- Due: **To be submitted later.**
- **Note: Look at your floor plan and the different rooms.
- **Type: IP addresses MUST be enough for all host as each host MUST have its own unique IP address.

TASK 6: DOCUMENTATION AND REFLECTION (10 Marks)

Now that you have designed your building and network, assigned all IP addresses to hosts, it is time to report all your work, and reflect upon them. In this task there are 3 important things to complete: a group report, an individual report and an individual reflection video.

The objective of the report is to report the work done and reflect upon how it can be made better (more that you are more "in the know"). You will be given a guideline for the reports but you are advised to expand it to give a clearer view of your work, your suggestions and your reflections.

Report format include: Times New Roman (12 font, 1.5 spacing, numbering for each task, A4 size paper, printed double sided (over the front)).

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MUHAMMAD AFIQ DANISH BIN MOHD HAZNI

MUHAMMAD SYAHMI FARIS BIN RUSLI A23CS0138

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